

Bathing Water Quality in Europe

(1990–2024)

Data Visualization • FMFI UK

Author: Samuel Pollák

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1. Introduction

Bathing water quality is one of the key environmental indicators, directly affecting public health, tourism development, and the state of ecosystems. The European Union has its own directive for bathing water quality (2006/7/EC), under which member states monitor hundreds to thousands of sites every year and assess their quality (European Commission 2006). The results are aggregated by the European Environment Agency (EEA), which publishes them as a publicly available dataset covering the period from 1990 onward.

The goal of this project is to explore this extensive dataset and answer four main questions:

- How has bathing water quality in Europe changed over the past 35 years?
- Are there significant differences between countries and between types of water (coastal, inland, transitional)?
- Is water quality related to a country's economic level, measured by gross domestic product?
- Does the intensity of coastal tourism affect the quality of coastal waters?

To answer these questions, in addition to the main EEA dataset we also used supplementary economic data from the World Bank (GDP per capita) and tourism data from Eurostat (number of nights spent in coastal regions). The results are presented through a set of visualizations along with a brief interpretation of the observed trends.

2. Data Used

In this project we worked with three datasets from public sources. The main dataset consists of annual bathing water quality assessments from the European Environment Agency, which we supplemented with economic indicators from the World Bank and tourism data from Eurostat. The original datasets had different formats and encodings, which required a series of steps to unify them before the actual analysis.

2.1 Bathing Water Quality (EEA)

The main dataset comes from the European Environment Agency (EEA 2024) and contains 672,629 records, where each row represents a single assessment of a specific bathing site in a given year. It covers 31 European countries over the period 1990–2024. The meaning of the individual columns is shown in Table 1.

Column	Meaning
countryCode	Two-letter country code (e.g. SK, IT, EL).
nameTxtInternational	International name of the site (beach, lake, river).
waterCategory	Type of water: coastal, inland, or transitional.
lon, lat	Geographic coordinates of the site (WGS84).
season	Year of assessment (bathing season).

quality	Quality category: Excellent, Good, Sufficient, Poor (the four-tier classification has applied since 2013; before 2013 the category Good or Sufficient was used, which we merged into Sufficient).
quality_score	Numeric score from 1 to 4 derived from the quality column (1 = Excellent, 4 = Poor), used to calculate averages and correlations.

Table 1: Meaning of the columns in the EEA bathing water quality dataset. The dataset contains 672,629 rows.

2.2 Gross Domestic Product (World Bank)

To compare water quality with economic level, we used GDP per capita data from the World Bank (World Bank 2024). We downloaded two versions: GDP in US dollars and GDP in purchasing power parity (PPP). The original dataset has one row per country and 71 columns, where each column represents one year. Before merging it with the other datasets, we converted the table into long format, so that each row contains a country, a year, and a GDP value.

2.3 Coastal Tourism (Eurostat)

To analyze the impact of tourism, we obtained a dataset from Eurostat (Eurostat 2024) on the number of nights spent at tourist accommodation establishments in the coastal regions of EU member states. The data is available from 2012 onward and contains annual aggregates in millions of nights for 27 countries.

2.4 Preprocessing and Merging

The most demanding step was unifying the country codes, since the World Bank uses three-letter ISO codes (e.g. DEU, GRC) while EEA and Eurostat use two-letter codes (e.g. DE, EL). Some countries also have inconsistent codes; for example, Greece is coded as GRC in the World Bank data but as EL in Eurostat. We handled these exceptions through manual mapping. After unifying the codes and aligning the formats, we merged all three datasets by the combination of countryCode and season into a single table containing 672,629 rows and 18 columns, which serves as the basis for all further analyses.

3. Bathing Water Quality in 2024

In the first part of the analysis we focused on the current state of bathing water quality in Europe in 2024. For each country we calculated the percentage distribution of assessments across the four categories (Excellent, Good, Sufficient, Poor) and ranked the countries in descending order by the share of beaches in the highest category (Figure 1).

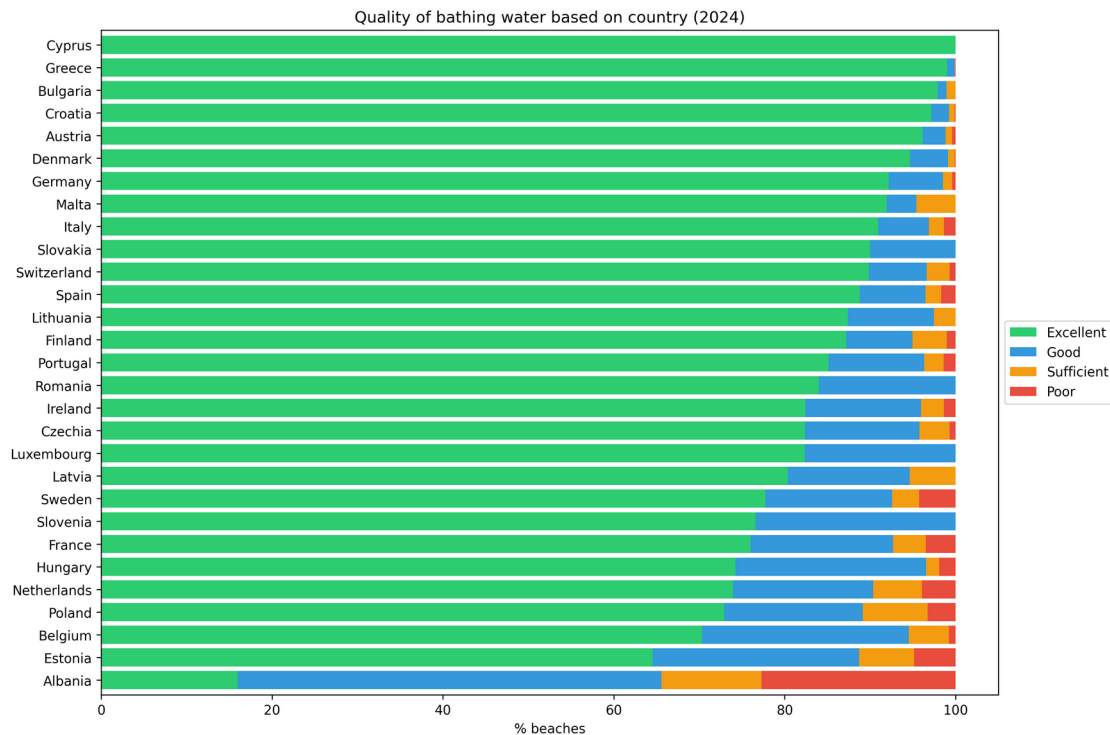


Fig. 1: Distribution of bathing waters by country according to quality category in 2024. Countries are ranked in descending order by the percentage share of beaches in the Excellent category.

Figure 1 shows a very high overall quality: for 12 countries, including Cyprus, Greece, Bulgaria, and Croatia, the share of the Excellent category exceeds 95%. The vast majority of countries have at least 80% of sites in this top category. A notable exception is Albania, where the share of the Excellent category reaches only about 15%, and more than 20% of sites fall into the Poor category.

The geographic distribution of quality is illustrated in Figure 2, which shows the share of beaches in the Excellent category for each country using a color scale. The map shows that high quality is not limited to the Mediterranean but also covers a large part of central and northern Europe. The lowest value among the countries studied belongs to Albania, which stands out in red.

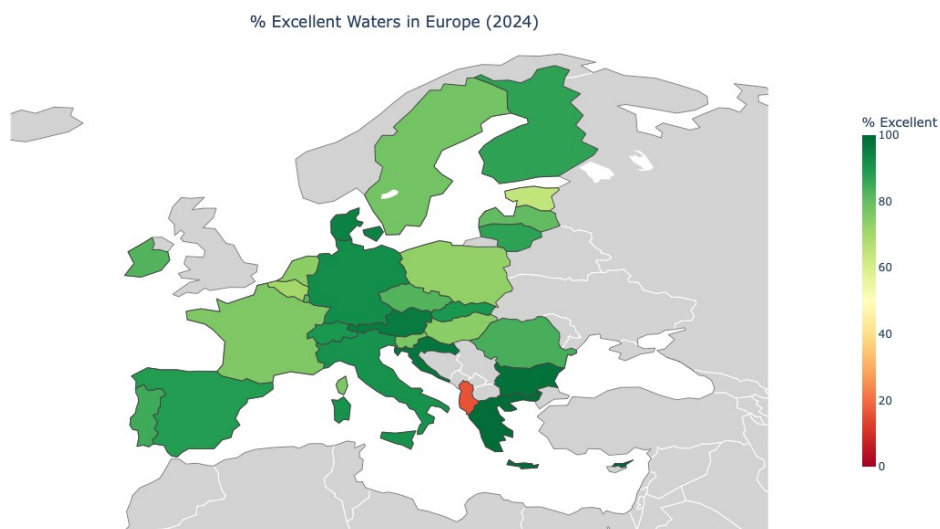


Fig. 2: Geographic distribution of the share of beaches in the Excellent category in 2024. Darker green corresponds to a higher share.

4. Water Quality Trends Over Time

The second analysis focused on developments over time. For each year from 1990 to 2024 we calculated the average water quality score across Europe (Figure 3). Recall that a lower score indicates better quality (1 = Excellent, 4 = Poor).

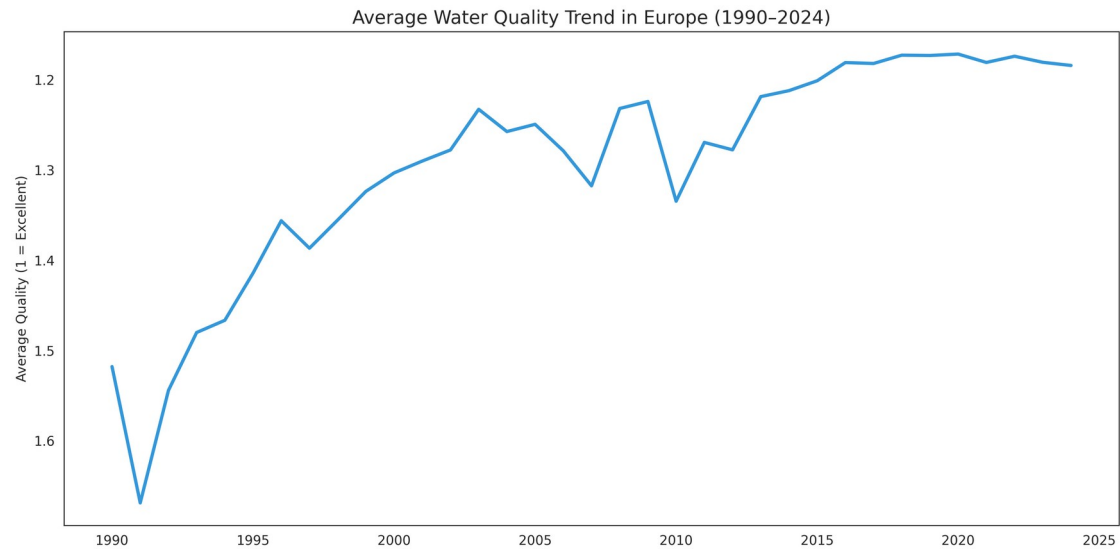


Fig. 3: Average bathing water quality score in Europe from 1990 to 2024. Lower values correspond to better quality.

Three phases can be distinguished in Figure 3. In 1990–1991 the average score reached its highest (worst) values, around 1.8. It should be noted, however, that only a few countries were part of the reporting at that time, so the overall average is quite sensitive to their specific values. Between 1992 and 2010 quality gradually improved with occasional fluctuations, which likely reflect one-off events in individual countries. Since 2011 the average score has settled at around 1.2 and has remained there ever since. This result suggests that bathing water quality in Europe has improved over the long term, although the data does not allow us to clearly determine the cause; it could be a combination of improved infrastructure, stricter monitoring, and the extension of EU legislation to new member states.

For a more detailed view we created a heatmap of the average score by country and year (Figure 4). The heatmap allows us to compare the trajectories of individual countries.

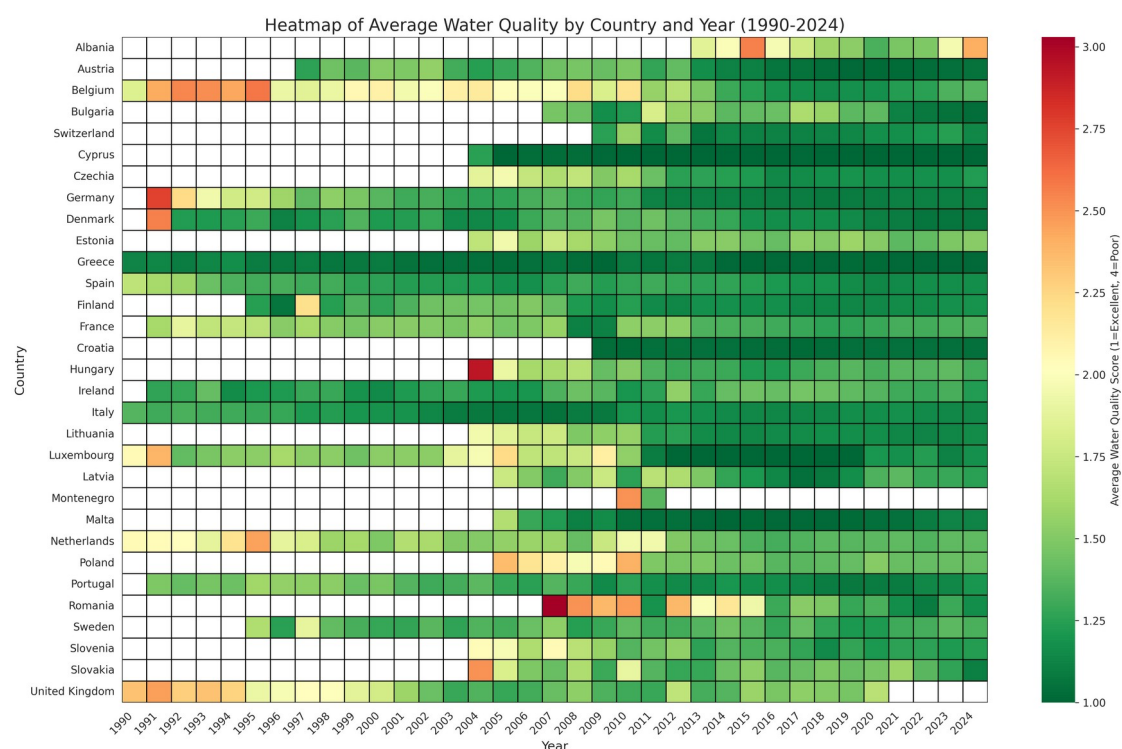


Fig. 4: Heatmap of the average water quality score by country and year, 1990–2024. Deep green corresponds to the Excellent category, red to the Poor category. White cells indicate that no data is available for that year.

Several interesting cases can be observed in Figure 4. In the 1990s Belgium recorded noticeably worse scores (orange cells), but it gradually joined the group of countries with high water quality. Romania shows a sharp spike in 2008 (dark red cell), followed by a gradual improvement. Greece and Cyprus have maintained top scores consistently throughout the entire period. Data for the United Kingdom is missing from 2022 onward, likely due to the country's withdrawal from EU reporting after Brexit (European Environment Agency 2024).

The spatial change across Europe as a whole is illustrated by comparing the years 2000 and 2024 in Figure 5. On the left-hand map, a large number of points in the Good or Sufficient and Poor categories (blue and red) can be seen, especially along the Atlantic coast and in the United Kingdom. On the right-hand map (2024), these points have almost disappeared and green dominates.

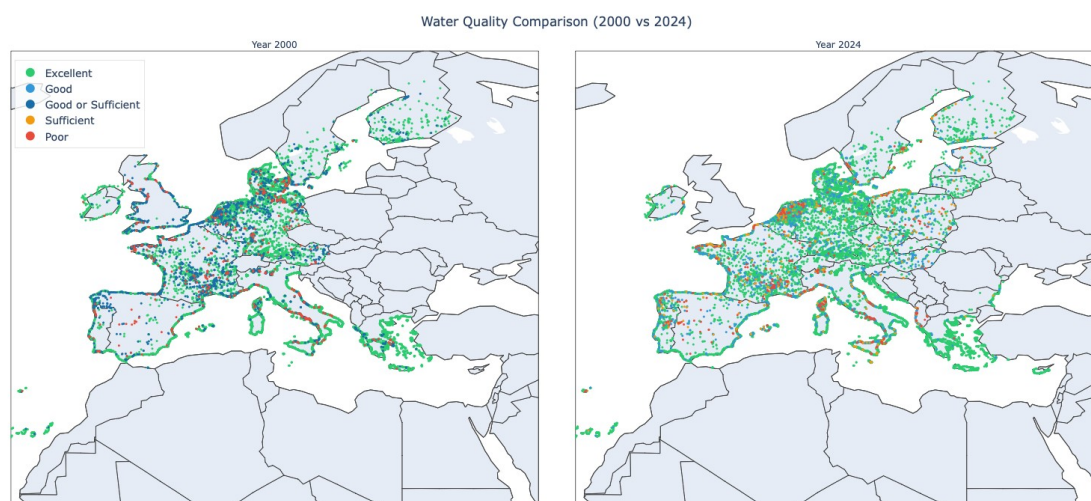


Fig. 5: Comparison of bathing water quality in Europe in 2000 and 2024. Each point represents one bathing site colored by its quality category.

5. Detailed View of Selected Countries

For a deeper analysis we selected four countries representing different types of situations: Spain (a combination of coastal and inland waters), Italy (extensive coastal as well as inland waters), Croatia (a shorter monitoring period, predominantly coastal waters), and Slovakia (exclusively inland waters, as our domestic context). Each country is shown separately in Figures 6–9.

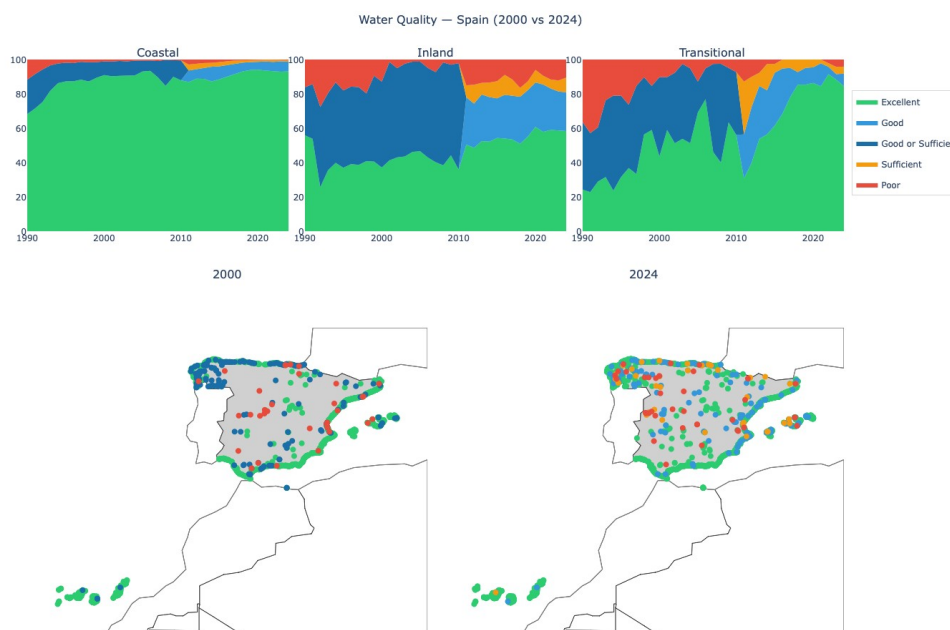


Fig. 6: Development of bathing water quality in Spain by water type (top) and spatial comparison of 2000 and 2024 (bottom).

Figure 6 shows that Spain's coastal waters were of high quality throughout the entire period. The most notable improvement took place in the northern coastal and transitional waters, where the share of the Poor and Good or Sufficient categories (red and blue at the top) decreased significantly over the last three decades.

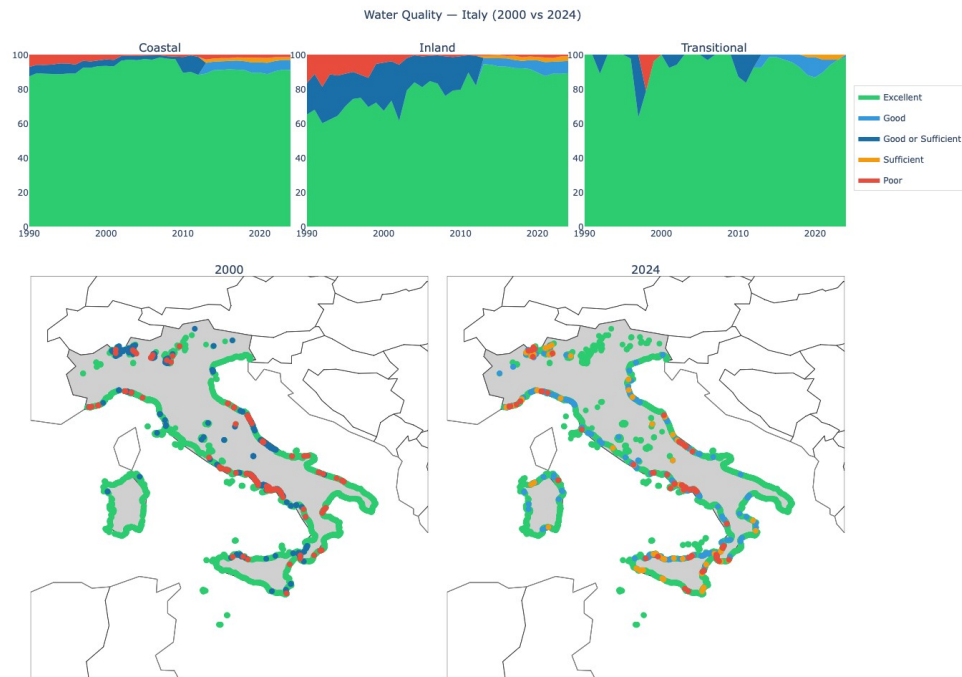


Fig. 7: Development of bathing water quality in Italy by water type (top) and spatial comparison of 2000 and 2024 (bottom).

Figure 7 shows that Italy's coastal waters were already excellent in 2000, with the entire coastline on the left-hand map shown in green. The most significant improvement occurred in the inland waters of northern Italy, where in 2000 there was a marked concentration of red and blue points. By 2024 the situation is much better, although some problem sites remain along the southern coast.

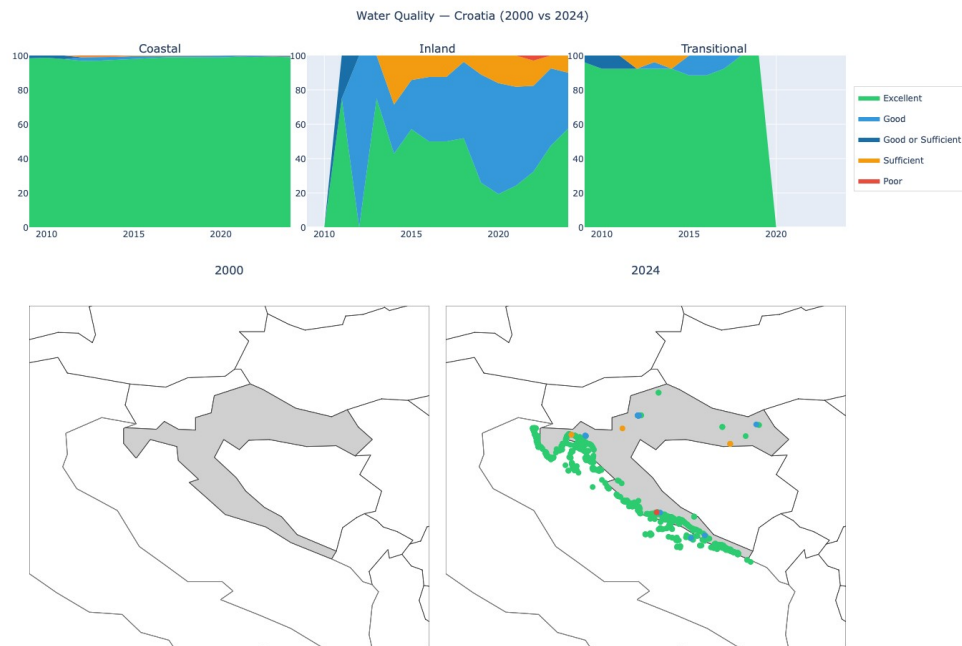


Fig. 8: Development of bathing water quality in Croatia by water type (top) and spatial comparison of 2000 and 2024 (bottom).

Croatia is almost a textbook example (Figure 8). Coastal waters reach practically 100% in the Excellent category throughout the period for which data exists. Monitoring in Croatia only began around 2010, so the left-hand map (2000) shows no data yet; by 2024, the sites on the right-hand map are overwhelmingly rated Excellent. Inland waters are somewhat more volatile within that same window. Transitional waters have been consistently excellent over the long term.

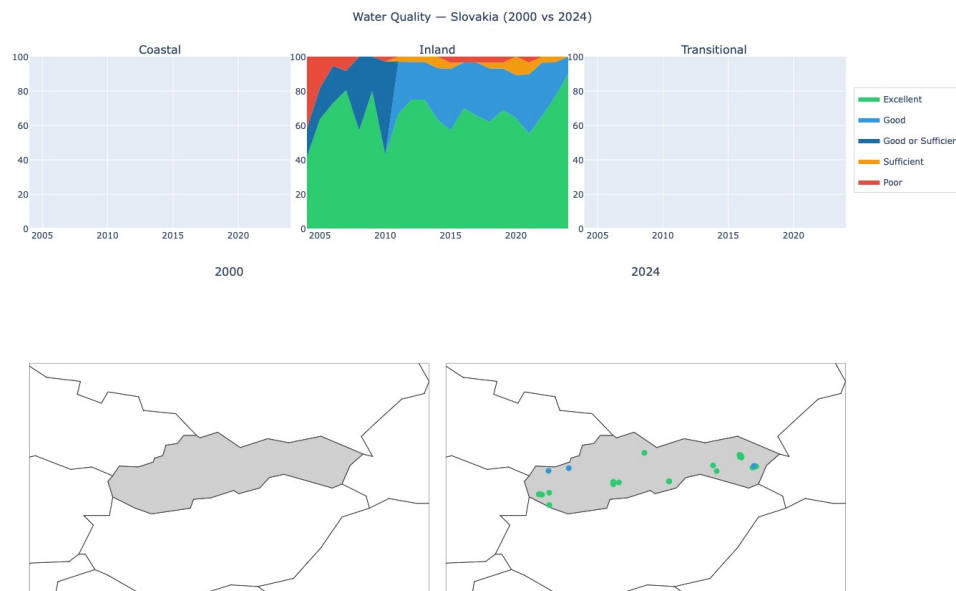


Fig. 9: Development of bathing water quality in Slovakia by water type (top) and spatial comparison of 2000 and 2024 (bottom).

Slovakia (Figure 9) is a special case, as it has neither coastal nor transitional waters, so the corresponding charts are empty. Monitoring data for Slovak inland waters is not yet available for the year 2000, which is why the left-hand map is empty; reporting effectively begins in the mid-2000s. By 2024, the inland sites shown on the right-hand map are rated Excellent or Good, with no Poor ratings remaining.

6. Comparison by Water Type

The next analysis examined whether quality differs by water type. For each year and type (coastal, inland, transitional) we calculated the percentage distribution of ratings across categories (Figure 10).

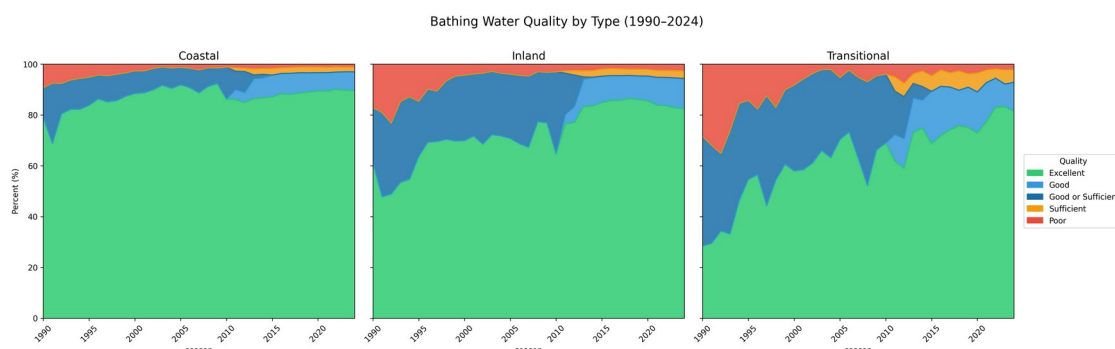


Fig. 10: Distribution of bathing water quality by water type, 1990–2024. Coastal waters (left), inland waters (center), transitional waters (right).

Figure 10 shows that coastal marine waters were the most stable throughout the entire period, with the share of the Excellent category always above 80%. Inland waters, meaning lakes and rivers, had the worst values in the early 1990s (around 50% Excellent), but gradually improved to a level comparable with coastal waters. The most dramatic improvement was recorded in transitional waters (river mouths), where the share of the Excellent category rose from about 30% in 1990 to 80% in 2024. This convergence in quality across all water types is consistent with the hypothesis that after the adoption of EU Directive 2006/7/EC, member states made targeted investments in wastewater treatment and the control of pollution from land-based sources.

7. Relationship with GDP

To examine whether a country's economic level is related to the quality of its waters, we constructed a scatter plot in which each point represents one country in 2024 (Figure 11). The x-axis shows GDP per capita, and the y-axis shows the average water quality score (lower is better).

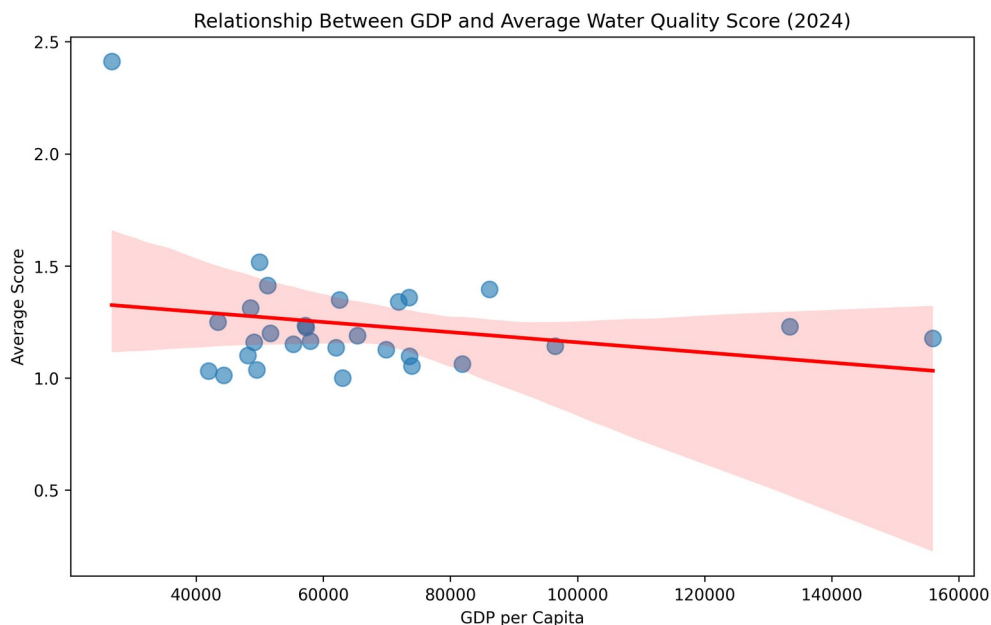


Fig. 11: Relationship between GDP per capita and the average water quality score for individual European countries in 2024. The red line represents the linear regression estimate, and the pink band its 95% confidence interval.

The regression line in Figure 11 slopes slightly downward, suggesting that wealthier countries have, on average, somewhat better water quality. However, the difference is very small and the confidence interval is fairly wide, so the correlation is weak. Our interpretation of this observation is that uniform European legislation (European Commission 2006) requires all member states to meet the same monitoring and treatment standards, so that differences in available financial resources translate less strongly into water quality than might be expected. It should be emphasized, however, that this is a hypothesis rather than a confirmed causal relationship. Other factors, such as geography, population density, or infrastructure investment made before EU accession, may also play a role.

8. Impact of Tourism on Coastal Waters

The final question we examined was whether the intensity of tourism has a negative effect on coastal water quality. For the years 2012–2024 we compared the annual number of nights spent in EU coastal regions with the average coastal water quality score (Figure 12).

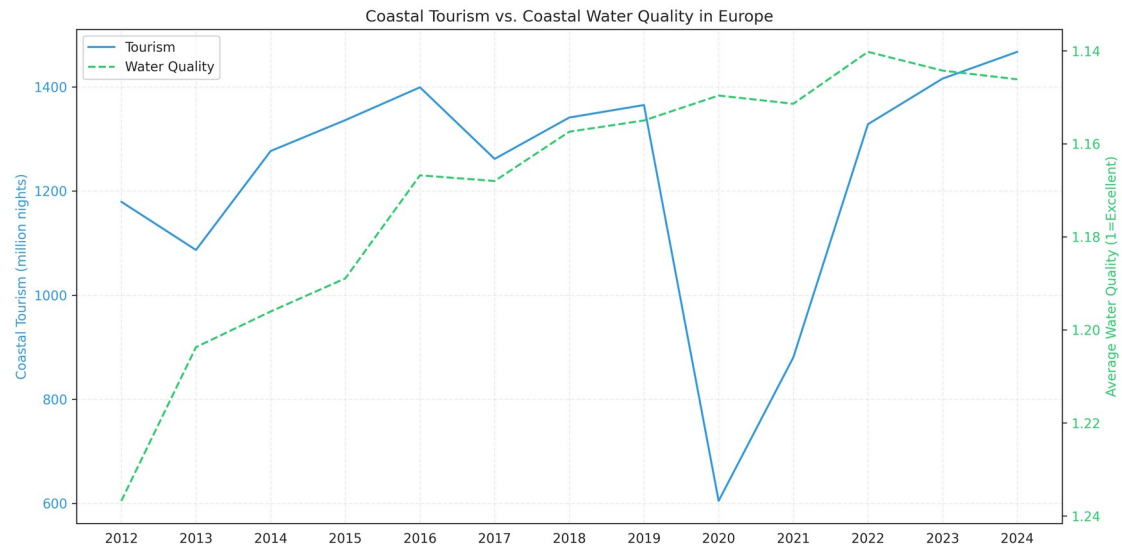


Fig. 12: Development of coastal tourism (blue, left axis) and average coastal water quality in Europe (green, right axis), 2012–2024.

Figure 12 shows two notable patterns. First, in 2020 coastal tourism fell by more than 50% due to the COVID-19 pandemic, yet water quality remained stable within its long-term trend. Second, in 2022–2024 tourism reached and then exceeded pre-pandemic levels, while water quality did not decline. These observations suggest that the direct relationship between the number of tourists and coastal water quality is weak in today's EU. This is likely related to the fact that tourist infrastructure in coastal regions includes wastewater treatment capacity sized for seasonal peaks. Caution is still warranted when drawing causal conclusions here as well, since the data covers a small number of observations (13 years) and many other factors, beyond the number of tourists, also affect water quality.

9. Conclusion

This project examined bathing water quality in Europe over the period 1990–2024 using three public datasets: EEA, World Bank, and Eurostat. The main findings can be summarized in three points.

- Bathing water quality in Europe has clearly improved over the past 35 years. The average score across Europe fell from around 1.8 in 1990 to 1.2 in 2024 and has remained at approximately the same level since 2011. This shift is visible across all three water types, with the most dramatic improvement seen in transitional waters (from 30% to 80% in the Excellent category).
- A country's economic level is not a decisive factor in the quality of its bathing waters. The correlation between GDP per capita and the average score is weak. A plausible hypothesis is the influence of uniform European legislation, which obliges all member states to meet similar standards regardless of their economic strength.

- The intensity of coastal tourism shows no visible relationship with a decline in coastal water quality. Even at record levels of tourism in 2022–2024, quality remained stable.

Several directions for further work are possible. It would be interesting to take a closer look at the case of Albania and understand why its quality is so markedly lower than that of other countries. The analysis could also be extended to additional countries outside the EU, or Europe could be compared with other regions of the world. It would be useful to add data on the quality of rivers and lakes that flow into the sea, in order to better understand the sources of pollution. It would also be interesting to look at data from before 1990, if available anywhere.

10. References

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